

CS 6440 Project Final Submission

Project/Application Name: SkinWise – Final Submission

Term: Spring 2024

Team Name: InnoHealth / Members: Jun Huang, Zhiqiu Jiang, Zifeng Zhang

Section I: Project Overview

Project Title: *SkinWise*

Team Members and Roles/Responsibilities (If Applicable)

- ☐ *Team Member 1 Jun Huang – Backend, Deployment*
- ☐ *Team Member 2 Zhiqiu Jiang – Frontend, User Experience*
- ☐ *Team Member 3 Zifeng Zhang – Machine Learning, User Interface*

TA Mentor: *Matthew Kalita*

External Mentor (If Applicable):

Project Task Status

(Insert your completed Task List as per the Canvas write up here.)

Completed Tasks
Work to form our team.
Team formation and project kick off. Submit Practicum sprint # 1.
Brainstorm team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.
Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.
Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.
Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.
Explore the patient resources from FHIR or available FHIR servers. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.

Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.
Testing and deployment of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.
Deployment of the SkinWise web application with CRUD functionality
Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.
Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.

Section II – Project Artifacts

Deployed Application URL: <http://ec2-13-60-53-97.eu-north-1.compute.amazonaws.com:8501>

(A link to your deployed project, Tableau dashboards, or anything else that exists remotely and acts as the user interface for your project.)

GitHub Repository Link: <https://github.gatech.edu/jhuang709/SkinWise/>

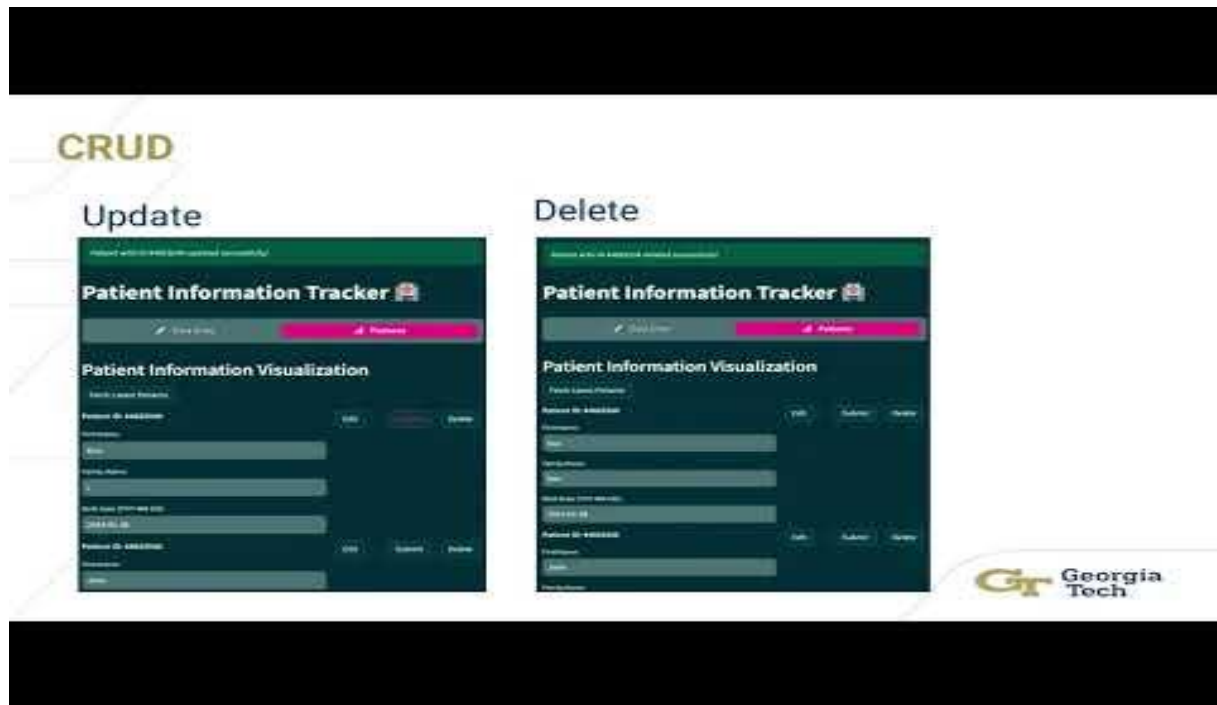
NOTE: All of your code and other project artifacts must be in this repository. This must be a Georgia Tech Enterprise Repository with TA mentors given full access.

GitHub Branch or Commit to Grade (If Applicable): *(If you have a specific commit or branch you would like us to grade instead of main/master, please state it here.)*

Section III – Project Presentation

Please see the Canvas Assignment page for the current semester for general guidelines on the presentation.

Presentation Link: <https://www.youtube.com/watch?v=dDfNKfDLWtM>



Note: Convert **Slides to PDF** and Merge with this single submission. You could add the contents of this deliverable to the final slide deck and convert the whole thing to a PDF to submit. Just make sure all links are clickable and do not include the submission information in the video presentation. (This is to provide a backup to presentations, both to ensure smoother grading in case of an error in a presentation and to maintain a record in Canvas.)

Section IV – Project Documentation

Create a directory in your GitHub repository called “Documentation”. Within this directory provide the following:

<https://github.gatech.edu/jhuang709/SkinWise/tree/main/Documentation>

□ Technical and/or User Manual

<https://github.gatech.edu/jhuang709/SkinWise/blob/main/Documentation/Technical%20and%20User%20Manual.md>

- Provide information on deployment of the application.
- Provide basic user instructions in order to utilize the application fully and show off all features.
- Special Instructions for Grading (“navigate to this URL...type in the following...click on this...etc.”) for the instructing team to follow in order to successfully deploy and run your application (in the event it is necessary, trust us it happens every semester). Include any special build or launch instructions. Please do not forget login information or anything such as FHIR patient IDs that may be needed.

- **PLEASE USE MARKDOWN SO NO DOWNLOAD/CLONE IS REQUIRED.**

- Research
 - https://github.gatech.edu/jhuang709/SkinWise/blob/main/Documentation/Team_SkinWise_IHI_Research.pdf
 - Any research done during the course of the project. This will vary heavily by project. As always, check with your TA mentor on expectations.
- Architecture Diagram
 - <https://github.gatech.edu/jhuang709/SkinWise/blob/main/Documentation/Architecture%20Diagram.png>
 - Include an up to date version of your architecture diagram from Sprint #2.
- Startup and Configuration Files
 - https://github.gatech.edu/jhuang709/SkinWise/tree/main/Documentation/Startup_and_Configuration_Files
 - Dockerfile, docker-compose.yml, etc..., as applicable

For each item included, provide a labeled direct links. Use full URLs, do not embed the link in a word or phrase.

All of the documentation that you submit with your final project should be treated as an official deliverable – this includes the technical/user manual(s). Documentation that is not well structured or clearly written may result in point loss, as it can impact use and thus grading of the application.